

The Hodgkin-Huxley Model

Its Extensions, Analysis and Numerics

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THE HODGKIN-HUXLEY MODEL

BACKGROUND INFORMATION

The Journal of physiology presented a series of papers in 1952 that would forever change the relationship between mathematics and physiology. Alan Lloyd Hodgkin and Andrew Huxley authored a succession of five papers describing the nonlinear ordinary differential equations that model how action potentials can be initiated and propagated through an axon [1-5]. This project will touch on the first paper in the series to give a background identity to the differential equation and its parameters and focus mainly on the fifth article which describes the mathematical model to be analyzed in future sections.

In 1963, the Nobel Prize for physiology and medicine was awarded to Hodgkin and Huxley for their ground-breaking research on the squid giant axon. The squid, with its tremendously large axon (up to 1mm diameter) and minimal conductance [6], allowed the pair to perform many experiments that have wide applications across organisms.

Cole and Curtis in 1939 were the first to identify the increase in membrane conductance during an action potential [7] following from a study in 1902 by Bernstein to correctly show the membrane separating two solutions of different ionic composition [8]. Bernstein studied the ionic composition of the solutions surrounding the membrane and proved there is a much greater concentration of potassium within the cell and an increased level of sodium outside. He used this information to conclude that the resting membrane potential is close to the potassium equilibrium value during rest, but rose to zero during activity. Bernstein reasoned that the activity caused the membrane to degrade, but this was proved incorrect in 1938.

During the year of 1938, Hodgkin collaborated with Cole to directly measure the membrane voltage [9]. This experiment was successful and they noticed that during activity the membrane potential will rise from the resting value (around -65mV) to well exceed the proposed value of zero (by Bernstein [8]) and

almost reverse sign at the peak (Figure 1). Hodgkin and Cole were able to attribute this rise (action potential upstroke) to the changing permeability of different ions.

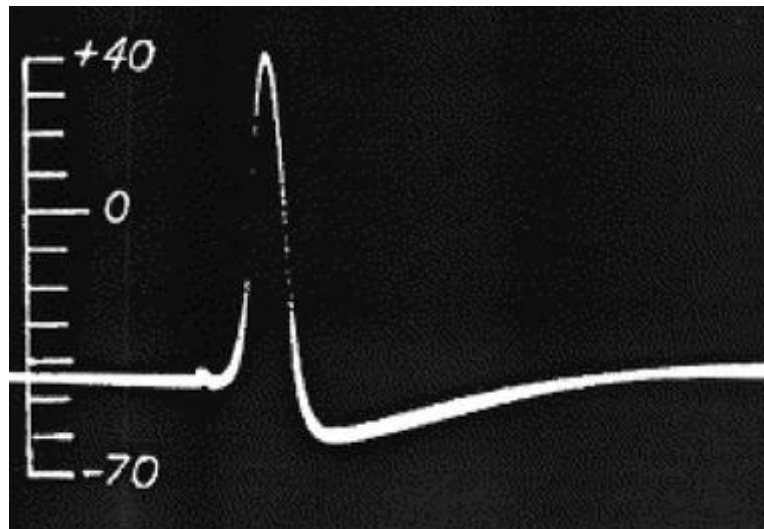


FIGURE 1 THE ORIGINAL DEPICTION OF A SINGLE ACTION POTENTIAL OVER TIME [9]

To further analyze the membrane changes, the voltage clamp experiment was created. Two pairs of electrodes read the voltage drop, while another pair injects current to clamp the voltage to a specific level. The changes in permeability and the dynamic link to membrane voltage were finally addressed and modeled in 1952 by Hodgkin and Huxley [1-5].

ANATOMY OF THE NEURON

Neurons receive signals through dendrites located on the main bulk of the cell. From there, the signal is propagated through the body and sent along the axon to the next neuron (Figure 2). Neurons send signals to other neurons via action potentials. An action potential is an explosion of depolarizing current that travels along the cell. For an action potential to occur, the depolarization must reach a minimum threshold voltage. Action potentials are only fired as an 'all-or-none' response. This means that action potentials do not vary in size and will not occur if threshold is not reached.

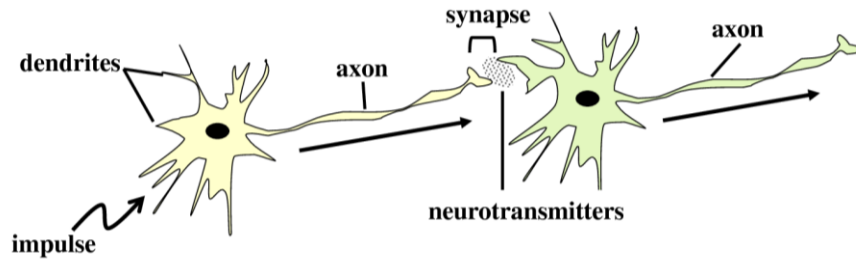


FIGURE 2 TWO NEURONS SIDE BY SIDE SHOWING THE PATH OF AN ACTION POTENTIAL. THE IMPULSE IS RECEIVED BY THE DENDRITES CLOSE TO THE BODY. IT THEN MAKES ITS WAY TO THE AXON WHERE IT TRAVELS TO THE NEXT NEURON OR ENDPOINT. THE SYNAPSE IS A LOCATION OF RELEASE OF NEUROTRANSMITTERS TO BEGIN THE ACTION POTENTIAL AT THE NEXT NEURON. TAKEN FROM A PUBLIC SOURCE.

ACTION POTENTIALS

Action potentials occur in excitable cells including neurons, muscle cells, and endocrine cells. In the case of neurons this is a signal that is propagated to relay information. The stepwise process of an action potential is described below.

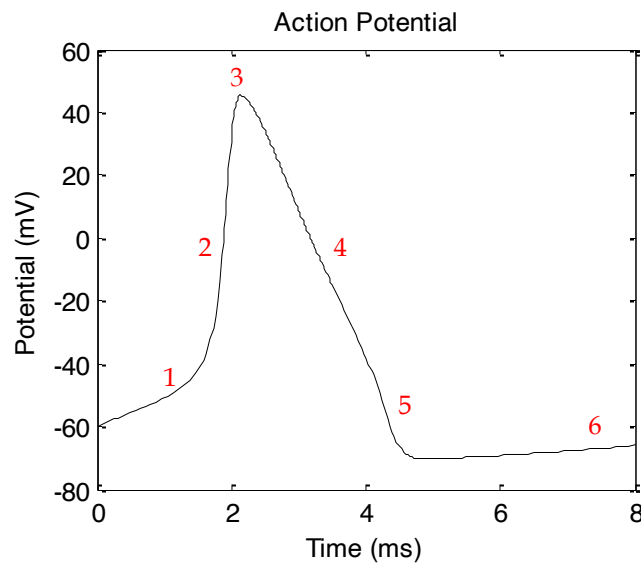


FIGURE 3 THIS IS AN ACTION POTENTIAL CREATED FROM THE MATLAB CODE PRODUCED LATER IN THIS PROJECT. EACH TIME POINT OF THE CURVE IS EXPLAINED IN THE TEXT BELOW.

Figure 3 above is a copy of the action potential created using the Matlab program to solve the Hodgkin-Huxley equation. Each part of the graph will be broken down according to the number sequence:

1. At this point in time, the firing threshold has been reached and the action potential begins to fire. The sodium channels are now open and an influx of sodium occurs.
2. The membrane becomes more positive at this point due to sodium entering the cell. Now, the potassium channels open and potassium leaves the cell.
3. At the peak of the action potential, the sodium channels have now become refractory and no more sodium is allowed to enter.
4. During the downfall phase of the action potential, the sodium channel is still in its refractory stage and only the potassium ions are passing through the membrane. As the potassium ions continue to leave the cell, the membrane potential moves toward the resting value.
5. At this point, the potassium channels close and the sodium channels begin to leave the refractory phase and reset to its resting phase.
6. The diffusion of extracellular potassium away from the cell causes a very slight increase in membrane voltage. It finally returns to its resting value where it can await another action potential. While the membrane is hyperpolarized (below resting) the cell cannot fire. This prevents the action potential from travelling backwards.

MATHEMATICAL MODEL

The Hodgkin-Huxley model is based on the parallel thought of a simple circuit with batteries, resistors and capacitors. A basic model of this circuit is shown in Figure 4. Current can be carried through the circuit as ions passing through the membrane (resistors) or by charging the capacitors of the membrane [5].

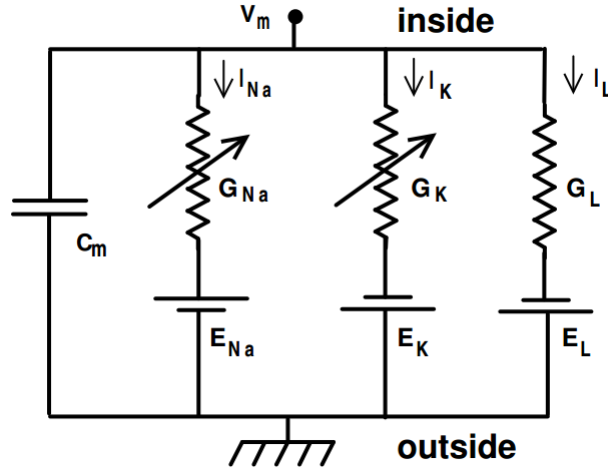


FIGURE 4 THIS FIGURE SHOWS THE MODEL OF THE CELL AND ITS FLOW OF CURRENT IN A SIMPLE CIRCUIT DIAGRAM. THE CAPACITOR OF THE MEMBRANE IS SHOWN ON THE FAR LEFT AND THE INDIVIDUAL IONIC CURRENT IS SEEN TO TRAVEL DOWN ITS RESPECTIVE BRANCH[5].

The electrical activity depicted in Figure 4 can be summarized as the famous Hodgkin-Huxley model:

$$C_m \frac{dV_m}{dt} + I_{ion} = I_{ext},$$

where C_m is the membrane capacitance, V_m is the intracellular potential, t is the time, I_{ion} is the net ionic current flowing across the membrane, and I_{ext} is the externally applied current [5]. The most significant advantage of the above equations is to determine the membrane capacitance in such a way that is independent of the sign or magnitude of the intracellular potential and minimally affected by the time course of V_m .

IONIC CURRENT

The ionic current described in Hodgkin and Huxley's giant squid axon can be divided into three different cases: sodium current, potassium current, and the leakage current [5]. The movement of each of these currents is proportional to the conductance times driving force [5]. For example, $I_k = G_k(V_m - E_k)$.

Conductance can easily be described by using the notion of the membrane channel. However, in 1952 when Hodgkin and Huxley were developing their model, there was very little knowledge about these ion

channels. The initial findings of the model properties still hold true for the additional information acquired about ion channels. For convenience, the 1952's Hodgkin Huxley model will be explained using today's knowledge of membrane ion channels.

The conductance is caused by the opening of many microscopic channels in the membrane. Each individual membrane contains many gates and when all gates are in the *permissive* state (allows ions to pass through) then the channel is considered to be *open* [6]. Once a channel is open, all ions of the specific channel can flow freely through. The probability of a gate being in the permissive state depends on the current value of the membrane voltage.

To develop the differential equations that describe the conductance, the probability of a gate being open is defined as p_i for any ion, i . Looking at the macroscopic scale, p_i can also be the fraction of gates that are in the permissive state. Also, the amount of gates closed are $(1 - p_i)$, by definition. To obey the first-order kinetics [6],

$$\frac{dp_i}{dt} = \alpha_i(V)(1 - p_i) - \beta_i(V)p_i,$$

where, α_i and β_i are the rate constants dependent on the voltage that describe the transient rates of permissive and non-permissive gates.

All gates must be permissive simultaneously for a channel to be considered open. When this occurs, the channel will contribute a miniscule amount of conductance on a large scale, otherwise it will not contribute anything. The macroscopic conductance depends on many channels being open, which depends on an even greater amount of gates being in the permissive state. This leads to the equation,

$$G_i = \bar{g}_i \prod_i p_i,$$

where, $\bar{g}_i$ is the normalized constant that determines the maximum conductance when all the channels are open [5, 6].

Hodgkin and Huxley labeled the probability of a gate being in the permissive state as the name of the respective gate. To conform to this method, p_i is exchanged for the name of the gates – m , n , or h . Each channel is defined to have a specific amount of these gates. For example, the sodium channel has three m gates and one h gate [5]. Using this notation and the above theory, the ionic currents can be summarized by,

$$I_{ion} = \bar{g}_{Na}m^3h(V_m - E_{Na}) - \bar{g}_Kn^4(V_m - E_K) - \bar{g}_l(V_m - E_l),$$

$$\frac{dn}{dt} = \alpha_n(V)(1 - n) - \beta_n(V)n,$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m,$$

$$\frac{dh}{dt} = \alpha_h(V)(1 - h) - \beta_h(V)h.$$

To completely describe the Hodgkin Huxley model, the only thing that remains is to define the six rate constants.

VOLTAGE CLAMP EXPERIMENTS

This section will describe the voltage clamp experiment for the potassium channels. Both the leakage and sodium channels follow the same methods to obtain similar results.

In the fifth paper of the series by Hodgkin and Huxley, they discovered that as they kept the voltage constant, the conductance of the channel increased to a steady value dependant on the initial voltage [5]. Their most significant finding was that the rate at which the conductance reached its maximum depended greatly on the clamped voltage. More specifically, at 20mV the half-maximum value occurs after 5 ms, but at 100mV the half-maximum value is reached after only 2 ms [5].

Hodgkin and Huxley were then able to find the time course of the step value n in the circumstance where the voltage is increased by a certain increment after the steady-state is achieved. Initially, the resting value of n is

$$n_{\infty}(0) = \frac{\alpha_n(0)}{\alpha_n(0) + \beta_n(0)}.$$

But as the voltage is clamped to a different voltage, V_c , the steady state becomes

$$n_{\infty}(V_c) = \frac{\alpha_n(V_c)}{\alpha_n(V_c) + \beta_n(V_c)}.$$

A solution can then be found mathematically to solve $\frac{dn}{dt} = \alpha_n(V)(1 - n) - \beta_n(V)n$ with the above restraints. This simple solution is an exponential of the form [6]

$$n(t) = n_{\infty}(V_c) - (n_{\infty}(V_c) - n_{\infty}(0))e^{-\frac{t}{\tau_n}},$$

where [6]

$$\tau_n(V_c) = \frac{1}{\alpha_n(V_c) + \beta_n(V_c)}.$$

After all this information, Hodgkin and Huxley still needed to determine the proper order and type of gates for each channel. They used a trial-and-error method to find the proper order of the gates to achieve the sigmoidal conductance found from previous results. Since the value of n is bound between zero and one, it first needed to be multiplied by the normalization constant $\bar{g}_K$. In the case of potassium, the best match is the n value brought to the power of four, such that

$$G_K = \bar{g}_K n^4.$$

The conductance for the sodium and leakage channels was obtained by the voltage clamp in the exact same manner. The results from this section help to form the desired rate constants.

RATE CONSTANTS

The values of $n_{\infty}(V_c)$ and $\tau_n(V_c)$ obtained above by fitting the conductance data will allow $\alpha_n(V_c)$ and $\beta_n(V_c)$ to be found by using the following relationships [5, 6]:

$$\alpha_n(V) = \frac{n_{\infty}(V)}{\tau_n(V)},$$

$$\beta_n(V) = \frac{1 - n_{\infty}(V)}{\tau_n(V)}.$$

In Figure 5, the plotted circles represent the experimentally acquired values of $n_{\infty}(V_c)$, $\tau_n(V_c)$, $\alpha_n(V_c)$, and $\beta_n(V_c)$ as a function of voltage. Hodgkin and Huxley then determined curves that fit this data to give the following expressions for the rate constants:

$$\alpha_n(V) = \frac{0.01(10 - V)}{\exp\left(\frac{10 - V}{10}\right) - 1},$$

$$\beta_n(V) = 0.125 \exp\left(\frac{-V}{80}\right).$$

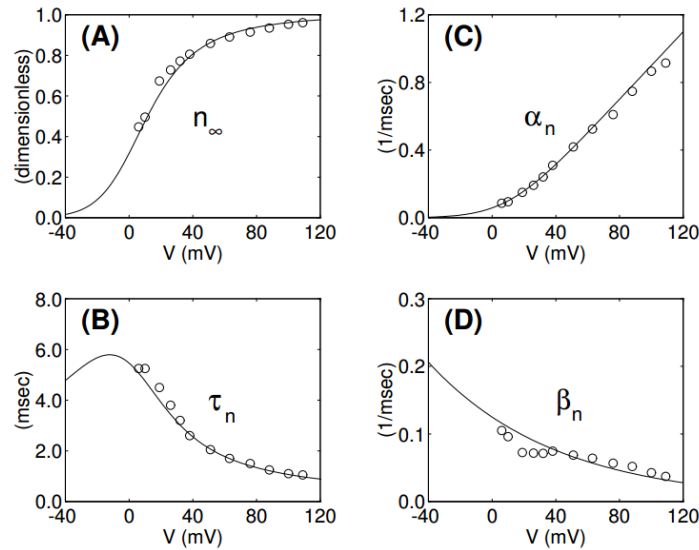


FIGURE 5 THE ASSOCIATED FUNCTION WITH THE N GATING VARIABLE. THE CIRCLES SHOW THE ACTUAL RECORDED VALUES WITH THE MODELED EQUATION AS THE SOLID LINE. (A) THE VALE OF N OVER 120MS. THE RATIO IS ALWAYS BETWEEN 0 AND 1. (B) THE TAU TIME VARIABLE. (C) SHOWS HOW ALPHA OF N CHANGES OVER TIME. (D) THE BETA VALUE OF N. [6]

To complete all the associated differential equations needed for the Hodgkin Huxley model the above methods are used to find the rate constants for the other gates. Thus,

$$\alpha_m(V) = \frac{0.1(25 - V)}{\exp\left(\frac{25 - V}{10}\right) - 1}$$

$$\alpha_h(V) = 0.07 \exp\left(\frac{-V}{20}\right)$$

$$\beta_m(V) = 4.0 \exp\left(\frac{-V}{18}\right)$$

$$\beta_h(V) = \frac{1}{\exp\left(\frac{30 - V}{10}\right) + 1}$$

That concludes all the necessary equations and constants that define the Hodgkin and Huxley model. The following sections deal with the different numerical methods used to solve the equations above.

CORRECTIONS, MODIFICATIONS AND EXTENSIONS

It has been sixty years since Hodgkin and Huxley made their ground-breaking discovery with minimal technology and background information. Since then, many have come forward to offer corrections and modifications to the model. There are many slight modifications to the original model to adjust to a different cell and different organism, but most follow the same original properties. For the remainder of this project, the original results will be respected, but listed below are some of the most common and obvious additions.

Hodgkin and Huxley showed the firing of an action potential with type 2 excitability. Type 2 excitability is sustained firing of action potentials from suprathreshold depolarizing current [10]. In 2008, John Clay et al. were able to separate this theoretical approach from experimental results by introducing a small modification to allow modeling of type 3 excitability. A simple modification provided in their 2008 paper slightly adjusts the potassium current [10].

There is an enormous amount of extensions to the Hodgkin-Huxley equation. Some examples are: bifurcations in the Hodgkin-Huxley model [11], speed dependent on frequency [12], and no external signal [13]. Many modifications to the model can be found by tweaking the results to match a different organism or different cell within an organism.

The Hodgkin-Huxley model can be summarized neatly into four separate ordinary differential equations with some supporting functions. These equations and their constants solved by the above methods are listed below [19]:

$$\frac{dv}{dt} = \frac{1}{C_m} \cdot [I - g_{Na}m^3h(v - E_{Na}) - g_Kn^4(v - E_K) - g_l(v - E_l)]$$

$$\frac{dn}{dt} = \alpha_n(v)(1 - n) - \beta_n(v)n$$

$$\frac{dm}{dt} = \alpha_m(v)(1 - m) - \beta_m(v)m$$

$$\frac{dh}{dt} = \alpha_h(v)(1 - h) - \beta_h(v)h$$

Where,

$$\alpha_n = \frac{0.01(v + 50)}{1 - \exp\left(\frac{-(v + 50)}{10}\right)}$$

$$\beta_m = 4.0 \exp(-0.0556(v + 60))$$

$$\alpha_h = 0.07 \exp(-0.05(v + 60))$$

$$\beta_n = 0.125 \exp\left(\frac{-(v + 60)}{80}\right)$$

$$\beta_h = \frac{1}{1 + \exp(-0.1(v + 30))}$$

$$\alpha_m = \frac{0.1(v + 35)}{1 - \exp\left(\frac{-(v + 35)}{10}\right)}$$

$$C_m = 0.01 \mu\text{F}/\text{cm}^2$$

$$g_{Na} = 1.2 \text{ mS}/\text{cm}^2$$

$$E_{Na} = 55.17 \text{ mV}$$

$$g_K = 0.36 \text{ mS}/\text{cm}^2$$

$$E_K = -72.14 \text{ mV}$$

$$g_l = 0.003 \text{ mS}/\text{cm}^2$$

$$E_l = -49.42 \text{ mV}$$

RESULTS

Solving the Hodgkin-Huxley model with numerical methods provided a solution familiar to the action potentials discussed in the introduction. All parts of the action potential are present including the rapid uprise, downfall and unexcitable phase. As can be seen in Figure 11, each different numerical method provides extremely similar action potentials. This is expected since the errors obtained for each method compared to the exact solution above were relatively small.

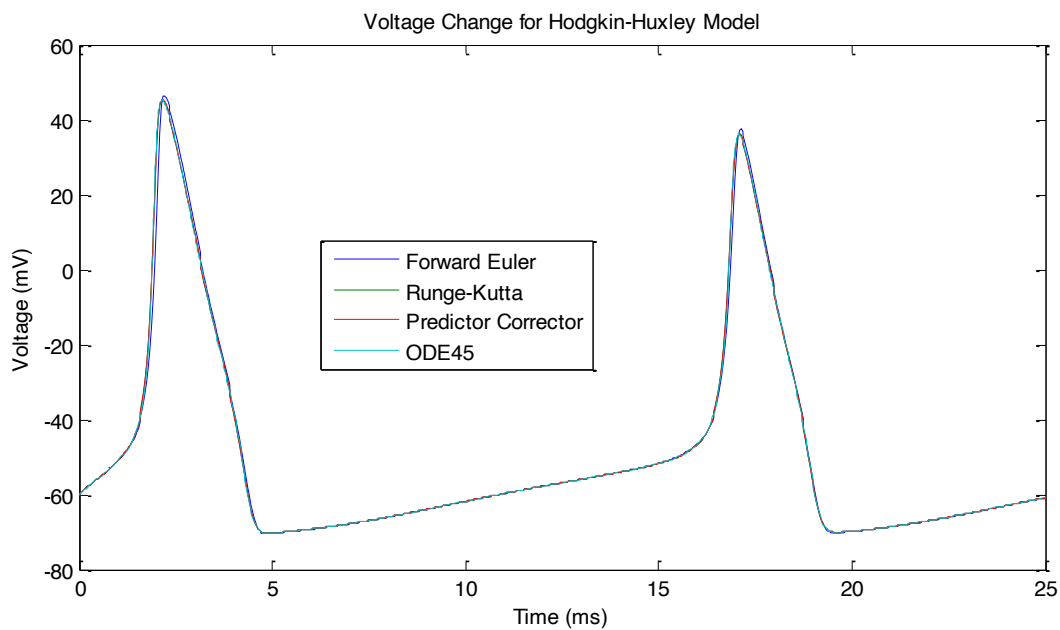


FIGURE 11 THE SOLUTION OF THE HODGKIN-HUXLEY MODEL BY FOUR DIFFERENT METHODS. TWO ACTION POTENTIALS ARE SEEN OVER THE 25SEC TIME SPAN.

The gating variables are correctly constrained between 0 and 1 in Figure 12. The variables are shown for five milliseconds but repeat in the same manner over longer periods of time. All methods are seen to have almost identical gating variables except for the ODE45 method.

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